GLOBAL NATURAL DISASTERS (2000–2025): HISTORICAL INSIGHTS, PER-CAPITA IMPACTS, AND FORECASTS WITH AN AUSTRALIAN FOCUS

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MINI PROJECT

Introduction

This project analyses global natural disasters from 2000–2025 using data from the EM-DAT International Disaster Database combined with population data for per-capita analysis.

The primary objectives were:

- To identify historical disaster trends by frequency, damage, and human impact.
- To assess per-capita disaster impacts across countries globally.
- To conduct a detailed case study on Australia, including historical patterns, per-capita impacts, and forecasting future disaster frequencies.

The dataset covers:

- 218 countries
- 6 disaster subgroups (Hydrological, Meteorological, Geophysical, Climatological, Biological, Extra-terrestrial)
- 2000–2025 disaster events, deaths, damages, and affected populations.

1. Data Overview

The dataset includes:

- Total Events: Number of disaster events recorded each year by country.
- Total Deaths: Mortality figures associated with disasters.
- Total Damage (USD): Economic damages adjusted to 2025 values.
- Population (2021): Used to derive per-capita metrics for better cross-country comparisons.

Key Summary Statistics

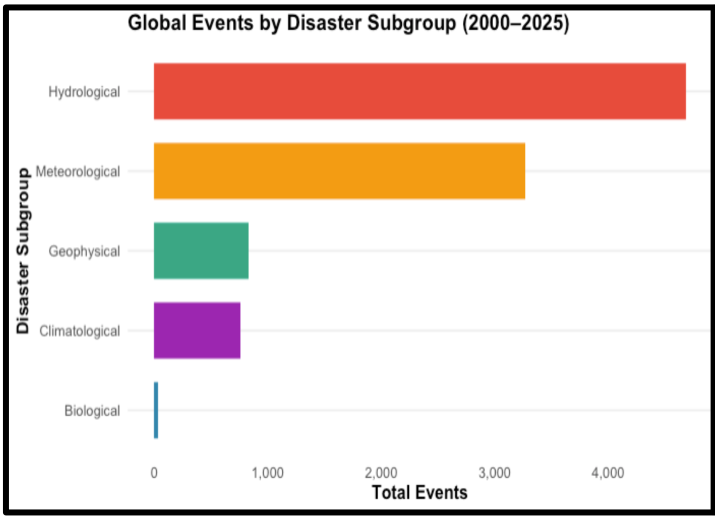
- Total Events: 9,575
- Total Deaths: 1.5M+
- Total Economic Damage: $4.9 Trillion

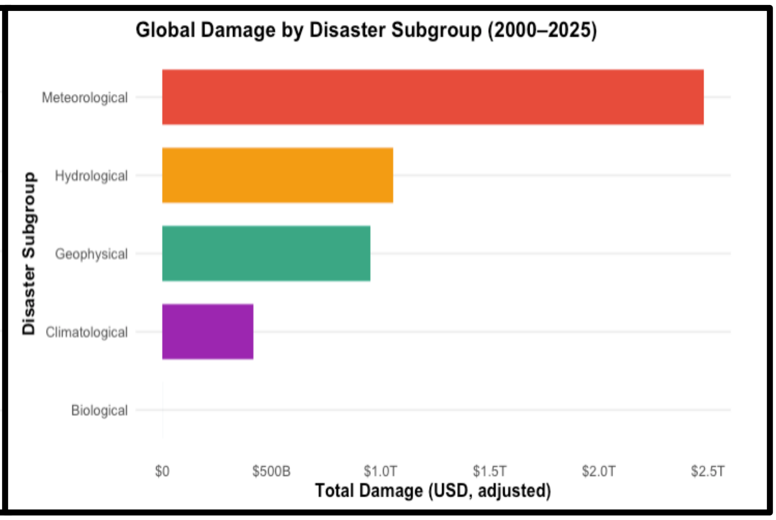
Years_Covered	Countries	Total_Events	Total_Deaths	Total_Damage
2000–2025	218	9575	1500203	4.902088e+12

2. Global Analysis

2.1 Disaster Distribution by Subgroup

Hydrological disasters (e.g., floods) dominate globally, contributing to ~49% of events but Meteorological disasters (e.g., cyclones) account for ~50% of total economic damages.

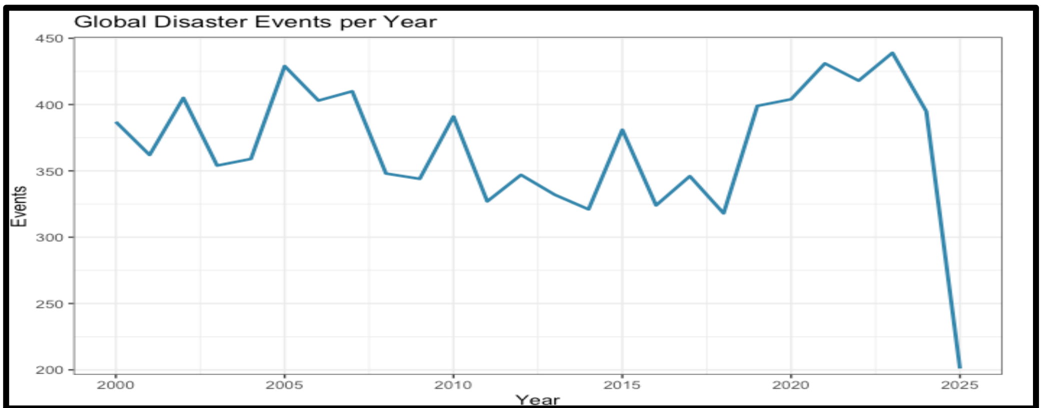


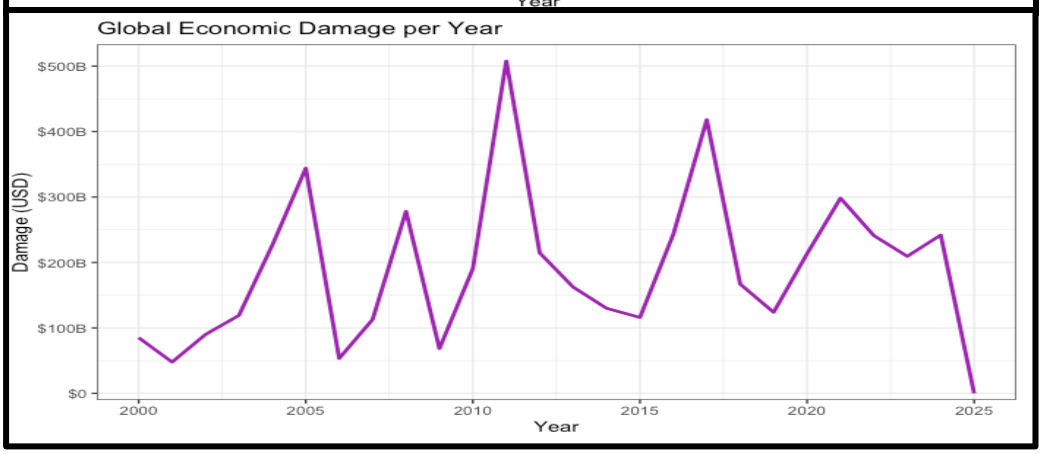


Disaster_Subroup	Events	Deaths	Damage	Events_Share	Damage_Share	Deaths_Share
Hydrological	4684	161364	1.058094e+12	48.92%	21.6%	10.8%
Meteorological	3273	513631	2.479197e+12	34.18%	50.6%	34.2%
Geophysical	830	798223	9.509869e+11	8.67%	19.4%	53.2%
Climatological	758	26973	4.135913e+11	7.92%	8.4%	1.8%
Biological	30	12	2.185985e+08	0.31%	0.0%	0.0%

2.2 Temporal Trends

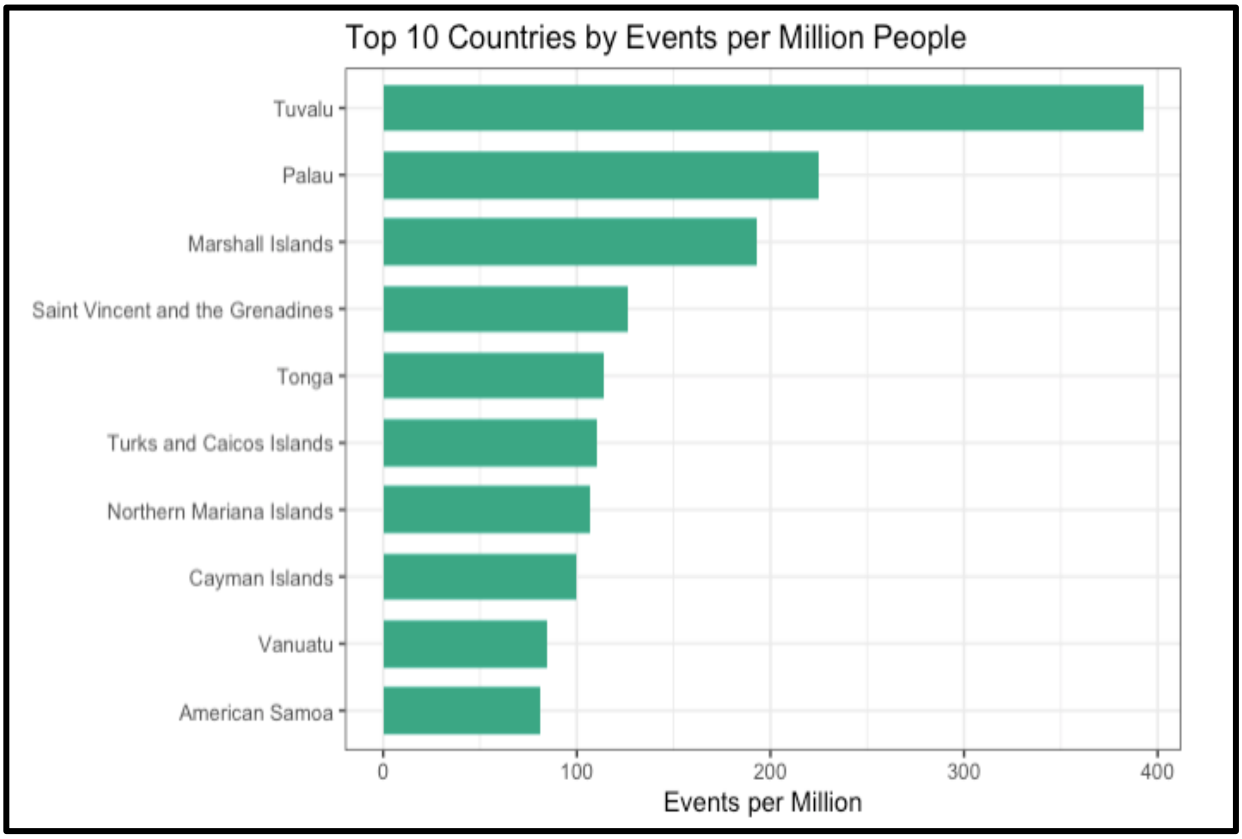
Disaster frequency peaked in the mid-2000s and early 2020s. Economic damages follow a highly volatile trend with major spikes during specific extreme events.





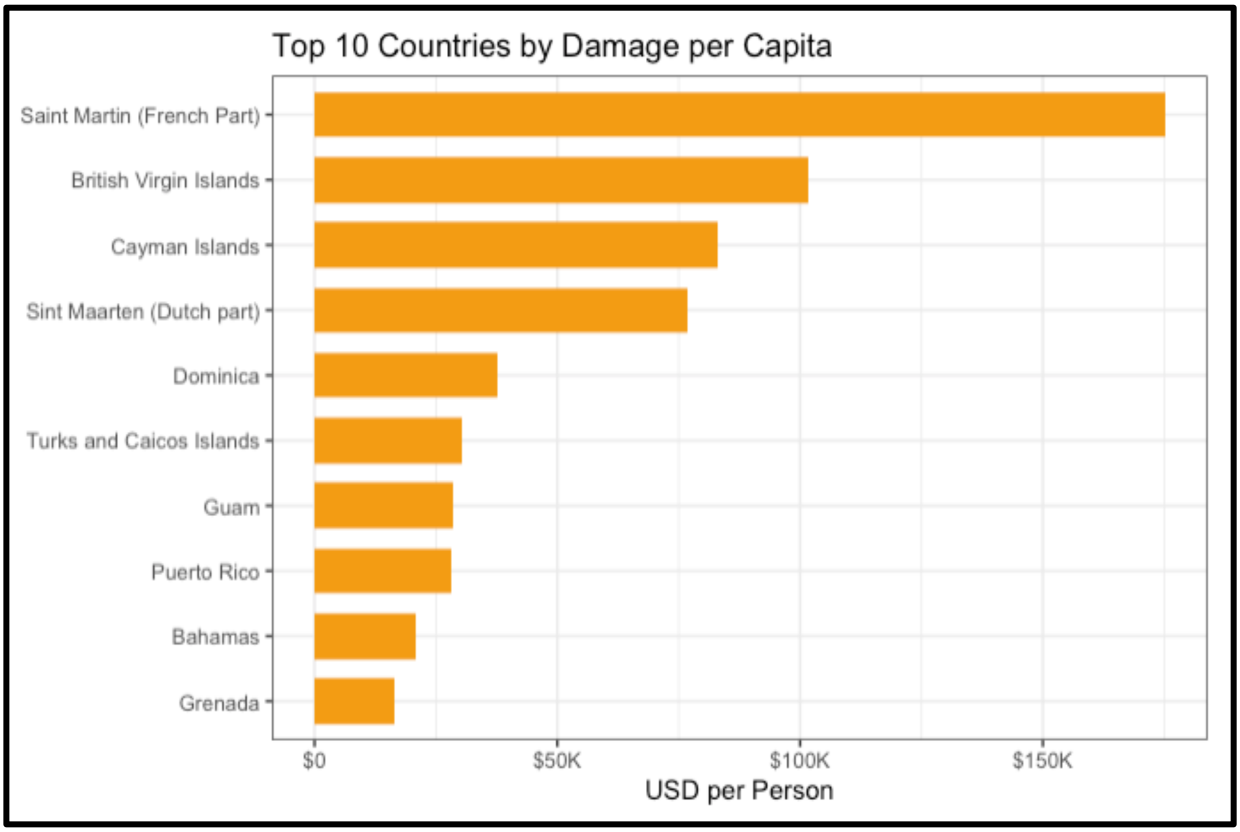
2.3 Per-capita Analysis

i) Top 10 countries by events per million people: Small Island nations dominate, led by Tuvalu (392 events per million) and Palau (225 events per million).



Rank_Events_per_Million	Country	Events_per_Million
1	**Tuvalu**	392.38768
2	**Palau**	224.93393
3	**Marshall Islands**	193.39087
4	**Saint Vincent and the Grenadines**	126.40873
5	**Tonga**	113.75486
6	**Turks and Caicos Islands**	110.51433
7	**Northern Mariana Islands**	106.43053
8	**Cayman Islands**	99.84595
9	**Vanuatu**	85.00399
10	**American Samoa**	81.25952

ii) Top 10 countries by damages per capita: The French Caribbean islands and British Virgin Islands recorded the highest economic damages per capita, exceeding $175,000 per person in some cases.



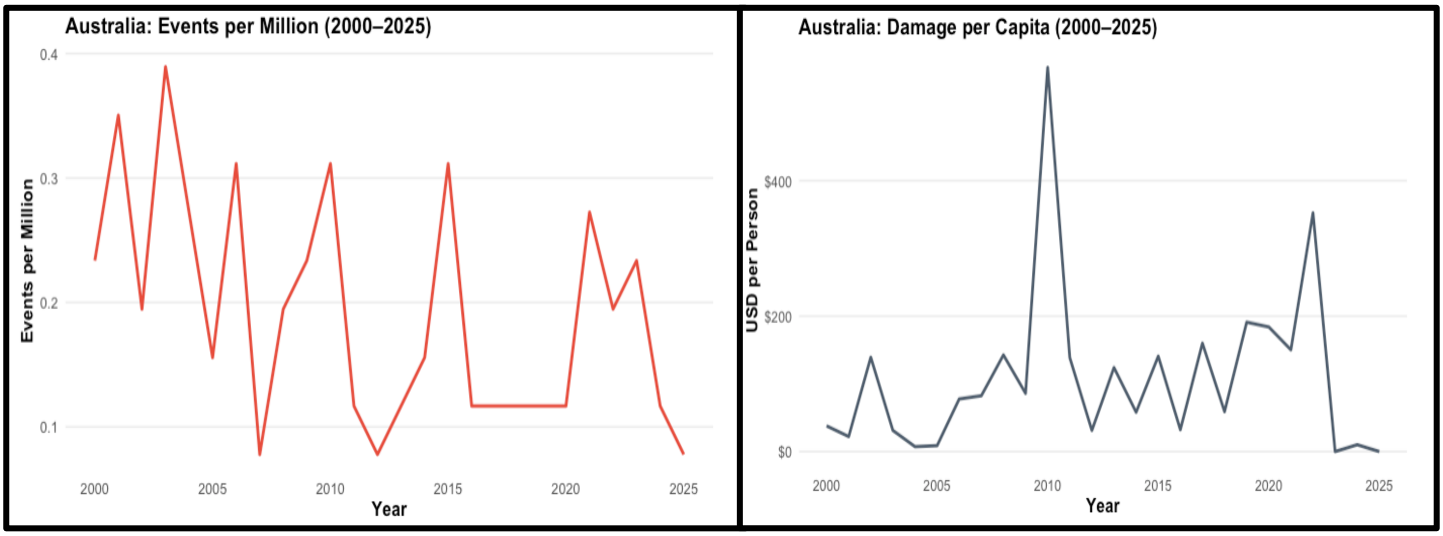
Rank_Damage_per_Capita	Country	Damage_per_Capita
1	**Saint Martin (French Part)**	175125.17
2	**British Virgin Islands**	101542.32
3	**Cayman Islands**	82925.45
4	**Sint Maarten (Dutch part)**	76960.98
5	**Dominica**	37685.62
6	**Turks and Caicos Islands**	30244.53
7	**Guam**	28432.18
8	**Puerto Rico**	28217.64
9	**Bahamas**	20939.01
10	**Grenada**	16337.68

3. Australia: Deep Dive

3.1 Historical Frequency and Damage

Australia contributed **~1% of global disaster events** but **<0.5% of economic damages**, indicating relatively fewer high-cost disasters compared to global averages.

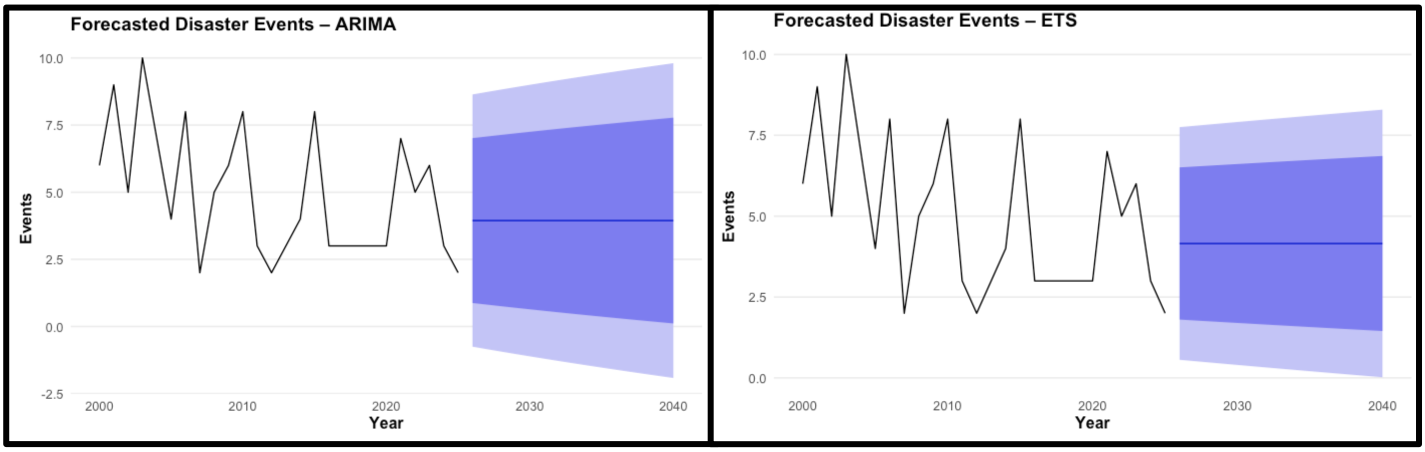
AU_Share_Events	AU_Share_Damage
1%	0%



3.2 Forecasting Disaster Frequencies

Using ARIMA and ETS models, I forecasted disaster frequencies for 2026–2040:

- Both models predict an average of 3–4 events annually, slightly lower than historical averages (4.9 events/year).
- Uncertainty intervals suggest potential variability, highlighting risk in long-term forecasting.



Model	Horizon_years	P80_High_First	P95_Low_First	P95_High_First
fc	15	7.013966	-0.7515473	8.639200
fc	15	6.505763	0.5603750	7.750065

Model	Horizon_years	Next_Year_Point	Mean_Point	P80_Low_First
fc	15	3.943826	3.943826	0.8736865
fc	15	4.155220	4.155220	1.8046772

3.3 Scenario Comparisons

Comparing 2000–2025 vs 2026–2040: Average annual events projected to **drop by ~50%** under baseline models.

Baseline_Events_avg_2000_2025	Scenario_Events_avg_2026_2040	Abs_Uplift_Events	Pct_Uplift_Events
4.923	2.438	-2.485	-50%

4. Statistical Insights

A simple regression analysis showed:

- A significant negative trend in disaster events over time ($p < 0.05$).
- $R^2 = 0.21$ suggests modest explanatory power, meaning temporal factors alone don't fully capture disaster variability.

term	estimate	std.error	statistic	p.value
(Intercept)	291.1452991	112.79817770	2.581117	0.01638830
Year	-0.1422222	0.05604839	-2.537490	0.01807604

r.squared	adj.r.squared	p.value
0.211534	0.1786813	0.01807604

5. Key Takeaways

a) Hydrological disasters dominate in frequency, but Meteorological disasters cause the most economic damage.
b) Small island nations face the highest per-capita risks due to geographic exposure
c) Australia:

- Historically low relative share in both events and damages.
- Forecasts suggest stable-to-declining frequency, but climate variability may cause unexpected shocks.

d) Global disaster damages remain highly volatile, emphasizing the need for resilient infrastructure and disaster risk financing.

6. Final Recommendations

A. Policy Focus: Prioritize climate adaptation in high-risk regions (e.g., Pacific islands).
B. Insurance & Reinsurance: Expand catastrophe risk pools for small nations.
C. Future Research: Integrate climate projections and socioeconomic variables for deeper risk modelling.

WEBSITES USED FOR DATA

https://databank.worldbank.org/embed/2021-Population-Data/id/67b1e32b
https://doc.emdat.be/docs/data-structure-and-content/emdat-public-table/